

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458723.91 +/- 0.017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.02 +/- 1.74 [%]
Orbital Inclination (inc)	87.54 +/- 2.92
Airmass coefficient 1 (a1)	0.44 +/- 0.29
Airmass coefficient 2 (a2)	0.47 +/- 0.22
Scatter in the residuals of the lightcurve fit is	76.13 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	6.75
Transit Duration (day)	0.086 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.05 vs 0.1404 (deviation 0.5σ)
Duration fit vs published	0.086 d vs 0.0754 d (deviation 0.49σ)
Mid-time O–C	2.8 min
Depth SNR	2.6
Residual scatter	76.13 %

Light curve

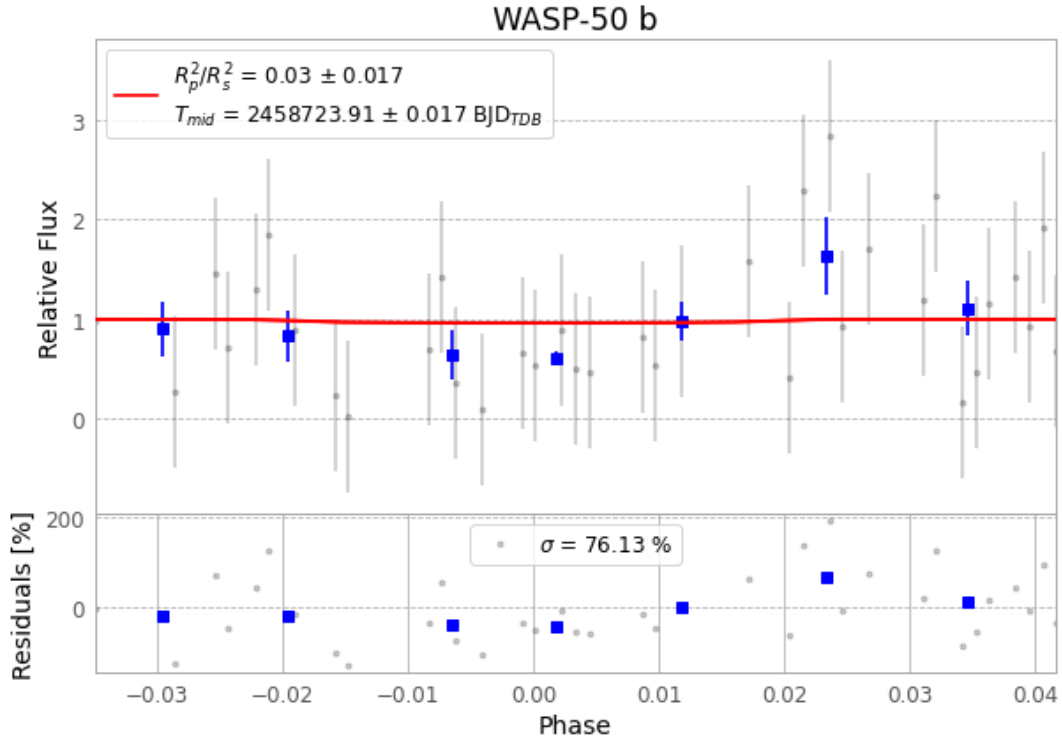


Figure 1 — FinalLightCurve_WASP-50 b_2019-08-28.png (SHA-256 eb999f5e6fde7c22...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a106c4833f2c9d22...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-50190828080312.FITS → WASP-50190828114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-190828.log (6945e3d7f863...), FinalLightCurve_WASP-50 b_2019-08-28.png (eb999f5e6fde...), FinalLightCurve_WASP-50 b_2019-08-28.csv (0920e85a75e4...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 01:38Z.